

Go-based Desktop App with **WAILS**

and *maybe* getting flagged by CrowdStrike in the process

@albertalrisa

Our story begins with...



Thread in #dev-glo...

Thread # dev-global-expansion



taishi Feb 12th at 3:09 PM

@Albert Richard Sanyoto Hi, Albert. I have a question.

The following file was detected as malware by EDR, but what does it do?

```
/Users/albert.sanyotorichard/Git/snkrdunk-en-ops/shell-t/bin/shell-t.dev.app/Contents/MacOS/shell-t
```

Good question, here's what it does



[en-ops.dev.app](https://en-ops.dev/app)

How did we end up with Wails

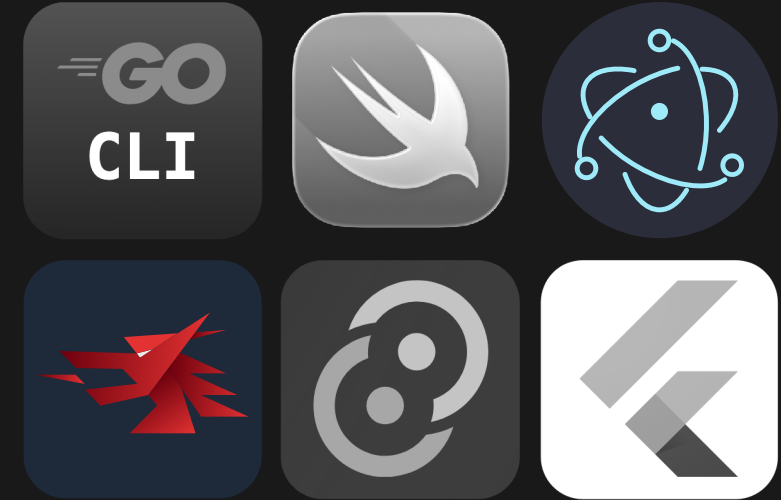
Requirements

User experience

Ease of development

Maintainability

(If possible) Code reuse



There are more Go-based UI frameworks but excluded due to it being:

- unmaintained (e.g. go-astilelectron),
- requires extensive development effort to create custom components with custom look and feel (e.g. Gio UI, using Raylib/SDL/Skia binding)

In comparison to Electron



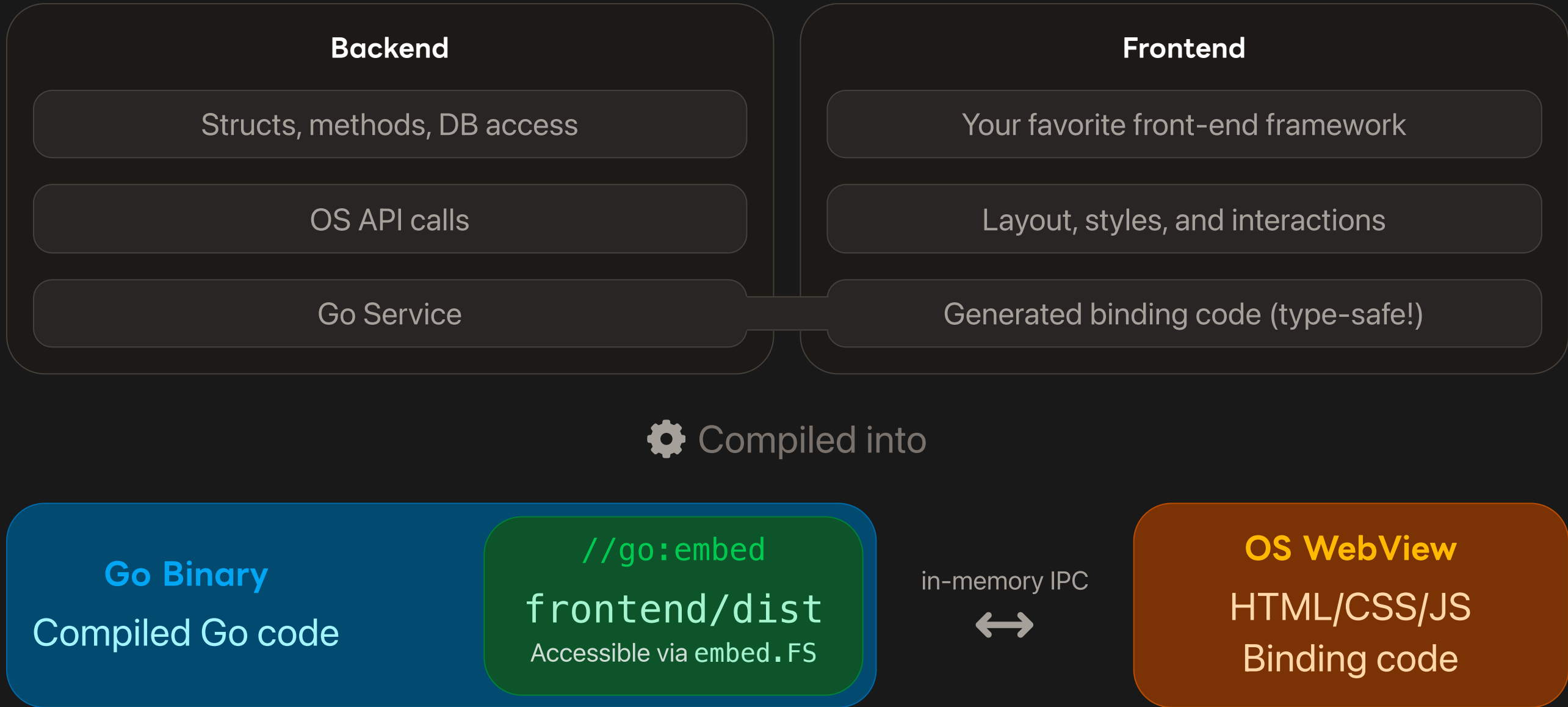
Electron



Wails

Language	JS or TS	Go
Browser	Bundled Chromium	OS-provided WebView WKWebView for macOS, WebView2 for Windows, WebKitGTK for Linux
Backend	Node.js	Go (compiled)
Bundle Size	min. 100MB	min. 20MB

What's in a Wails Application



Go ↔ TypeScript Bindings

Service code (Go)

```
type Demo struct{}

type Input struct {
    To      string
    Message string
}

type Output struct {
    OK      bool   `json:"ok"`
    Status  string `json:"status"`
}

func (s *Demo) SendRequest(in Input) (Output, error) {
    return Output{
        OK:      true,
        Status:  "pending",
    }, nil
}
```

If a service method returns (**value**, **error**), returning an **error** will result in a **Promise** rejection in JS/TS.

Generated bindings (TS)

```
export function SendRequest($in: $models.Input):
    return $Call.ByID(602180076, $in).then(($result) => {
        return $$createType0($result);
    });
}
```

```
export class Input {
    "To": string;
    "Message": string;

    /** omitted **/
}

export class Output {
    "ok": boolean;
    "status": string;

    /** omitted **/
}
```


Summary

Powered by



Use your favorite
frontend framework

CSS styling
(Tailwind support too)

Automatic binding
generation

Type-safe Go and
TypeScript

Two-way IPC
with stream support

Cross-platform

Auto-update
support

Might land you a
message from CIT
due to  **CROWDSTRIKE** alert

Thank you!



This slide (and its source code) can be found at
<https://github.com/albertalrisa/20260608-sodago-wails>